

Reproduction of omihub777/ViT-CIFAR: ViT from scratch on CIFAR-10

K-Veritas Team

Submitted by K-Veritas Team

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Independent reproduction · Artifact evaluation · Computer vision

Abstract

Trained the repo's 6.3M-parameter Vision Transformer from scratch on CIFAR-10 with the README command (200 epochs, Adam lr 1e-3, 5 warmup epochs then cosine, label smoothing, AutoAugment, 16-bit mixed precision, seed 42), logging with the CSV logger since no Comet key is used. Compatibility changes for current Python and PyTorch: pytorch-lightning 1.5.10 instead of the pinned 1.2.1, np.int replaced by int in autoaugment.py, and the per-epoch lr log call given on_epoch=True. Validation accuracy each epoch from the logged metrics. CIFAR-10 downloaded by torchvision. Single NVIDIA RTX 5060 Ti 16 GB, PyTorch 2.11 cu128, Python 3.12.

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	ViT CIFAR-10 (200 epochs)	2026-10-07	1	1	Signed by the K-Veritas server

2 Results

r1: ViT CIFAR-10 (200 epochs)

Run	Metric	Final value	Step
1	val_acc	0.9035	199
1	val_loss	0.777475	199

Run	Inline claim	Value	Phase
1	final_val_acc	0.9035	
1	best_val_acc	0.9043	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	3699.2 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	4173299f186e162faa0c74a8bd541a5602fc3fb8322eee07c093e46240d50431	8.1 MB
r1.kvbundle.zip	fa194f38343ff664c9c71fbc4a149952fd2a49c76be10d7e80c1f886d3ece655	16 KB

5 Links and licence

Paper: <https://arxiv.org/abs/2010.11929>

Reproduction of: <https://github.com/omihub777/ViT-CIFAR/tree/ab9043ea44cfe8311d30295fc0e0467d676187f1>

Record: CC-BY 4.0. Code: MIT, as declared by the authors. Tags: vision-transformer, cifar-10, image-classification.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (ViT CIFAR-10 (200 epochs)), kv:2610.00002v1/r1. Data hash 8ad214670a49a524d44d306473e42acf0853b332774cc38bf3a255843ccbf792.
<https://kveritas.org/records/2610.00002v1/r1.pdf>